



LambdaProof Application Module Definitions

1. DEVICE MODULE

Manages QIoT and compute devices participating in LambdaProof's lawful recursion cycle.

Purpose

To monitor, configure, and ensure that all registered devices (IoT, edge nodes, firmware gateways, sensors, signature modules) operate under:

- **prime-gated activation,**
- **zero-emission until lawful,**
- **CSL ethical commutation,**
- **drift-bound telemetry,**
- **and proof-carrying firmware events.**

Device Module — Screens

A. Device List Screen

Layout

- Table or grid with:
 - Device Name
 - Status (Active / Silent / Awaiting Proof / Offline)
 - Last Prime Epoch
 - Drift $\Delta\psi$
 - R96 value
 - Firmware ProofHash
 - Actions (View / Configure)

Functional Items

- Sort/filter by:
 - Prime epoch alignment
 - R96 health
 - Device type
 - Proof status
- Bulk actions:
 - “Enforce Silent Mode”
 - “Re-evaluate Devices”
 - “Generate Attestations”

B. Device Detail Screen

Sections

1. Device Overview

- Device type (QIoT, ZK sensor, Gateway, PrimeClock)
- Current prime epoch
- Activation schedule
- Last lawful transition
- $\Delta\psi$ drift graph
- R96 credits meter

2. Proof Attestation Pane

- Latest firmware proof
- Status: Valid / Expired / DriftExceeded
- proofHash, vkHash
- “Re-verify locally” button
- “Export proof” button

3. Prime-Gated Activation Panel

- Next activation window
- Required prime thresholds
- “Simulate next activation” (runs Policy Oracle in simulate mode)

4. Logs & Receipts

- Device-specific Archivum receipts
- Sorted by prime index
- “Export all receipts (zip)”

5. Controls

- Toggle: *Silent Mode ON/OFF*
- Trigger firmware proof generation
- Reset $\Delta\psi$ drift
- Re-bind device identity to MTPI root

Invariants

- No raw telemetry is ever sent. Only hashed commitments or ZK proofs.
 - Device cannot be activated outside a prime epoch.
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2. GOVERNANCE MODULE

Handles constitutional governance of MTPI/CSL rules, MIPs (MTPI Improvement Proposals), policy evaluations, and consensus outcomes.

Purpose

To expose the Λ Proof governance layer while respecting:

- CSL ethics
 - MTPI constitutional immutability
 - community-driven improvements
 - policy oracle constraints
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Governance Module — Screens

A. Governance Dashboard

Layout

- Governance summary
 - Active MIPs
 - Pending proposals
 - Last CSL verdict
 - Current prime epoch
 - Policy Oracle status
 - On-chain root hash of governance parameters
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B. MIP List Screen (MTPI Improvement Proposals)

Shows:

- Proposal title
- Status (Draft / Under Review / Open for Comment / Ratified / Rejected / Deprecated)
- Author
- Vote breakdown
- Last modified
- Required prime epoch for activation

Actions

- Create new MIP
- Comment on MIP
- View change history
- Simulate with policy oracle

C. MIP Detail Screen

Sections

1. Summary

- Intent
- Problem statement
- Motivation

2. Technical Spec

- Rule deltas
- CSL implications
- New drift bounds
- New prime-gating logic
- Verifier updates

3. Simulation Panel

Upload a proof or enter parameters:

- See whether the MIP passes or blocks the action.
- View effect on R96, $\Delta\psi$, CSL commutation.

4. Voting

- Vote interface (if user is authorized)
- Prime-indexed voting windows
- Result thresholds

5. Audit / History

- Original MIP text
 - All revisions
 - Provenance chain
 - Archivum entries for changes
-

D. Constitutional Layer Viewer

Visualizes the **CSL operators**:

- Neutrality
- Beneficence
- Silence Clause
- Commutation rules

Allows:

- exploring transformation graphs
 - running an action through CSL to see operator-by-operator outcomes
 - comparing past versions of the constitutional tensor
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3. DEVELOPER MODULE

Everything developers need to work with Λ Proof:

- circuits
- WASM/zkey artifacts
- proof builder APIs

- contract addresses
- test tools
- RPC profiles

Developer Module — Screens

A. Developer Overview

Includes:

- Network info
 - Contract addresses (MTPI_Core, Verifier, Poseidon)
 - RPC profiles (Default / DIN / Fallback)
 - Artifact versions available
 - Links to GitHub/Docs
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B. WASM / zk-SNARK Artifact Manager

Uploads & manages:

- `.wasm`
- `*.zkey.enc` (encrypted)
- `vk.json`

Tools

- Validate artifact hashes

- Export artifact bundle
- Regenerate VK hash
- Compare artifact versions
- Show dependency graph (root_contract / recovery_contract / MillerRabin)

Rules

- System must block raw `.zkey` upload unless encrypted
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C. Proof Builder (Developer Mode)

This version exposes:

- raw witness builder
 - arbitrary circuit selection
 - “Open WASM” debugging
 - watch-mode for dev circuits
 - publicSignals inspector
 - gas estimate for transition
 - ability to run proof only (no submission)
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D. Dev RPC / DIN Configuration

Includes:

- `RPC_URL_DEFAULT`
- `RPC_URL_DIN`

- RPC_URL_FALLBACK
- infraHealth panel
- latency matrix
- restaked trust score (DIN)

Actions:

- Test all RPC endpoints
- Auto-select best RPC route
- Export RPC profile for CLI/tooling

E. CLI Tools Page

Shows:

- install instructions

usage examples:

```
lproof proof build --artifact root --inputs inputs.json
```

```
lproof proof verify --vk root.vk.json
```

```
lproof proof submit --mode submit
```

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- test files
- template circuits

4. SETTINGS MODULE

Controls user, privacy, security, network, and advanced MTPI parameters.

Purpose

Maintain privacy guarantees, configure RPC profiles, manage keys, and set app-level behaviors.

Settings Module — Screens

A. User & Identity Settings

- Local user identity commitment
 - Key management
 - generate
 - backup (encrypted only)
 - restore
 - Show linked devices
 - “Regenerate identity commitment” (with warnings)
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B. Privacy Settings

- Toggle what metadata is allowed to leave the device
 - Default: **Zero-surveillance mode**
 - Option to opt out of sending RPC profile or client hints
 - Proof caching settings
 - Data retention window for local logs
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C. Network Settings

- RPC profile selector
 - Test connection button
 - Active DIN status
 - Auto-fallback rules
 - Graph: RPC latency & health
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D. Prime / Drift Settings (Advanced)

(For advanced users, researchers, or developers.)

Shows:

- current drift $\delta(t)$
 - limit (0.3Ξ)
 - R96 meter
 - $\Delta\psi$ change graph
 - “Request fresh prime evaluation”
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E. Local Storage & Security

- Manage IndexedDB / local storage
 - Clear proofs (safe wipe)
 - Enforce environment checks (secure context, https)
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F. App Diagnostics

- Version
 - Artifact versions
 - Proof engine logs
 - RPC logs
 - Local environment report
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Summary Table

Module	Purpose	Key Screens
Device	Manage QIoT & proof-carrying devices	List, Detail, Proofs, Activation, Receipts
Governance	Manage MTPI/CSL policy & MIPs	Dashboard, MIP List, MIP Detail, CSL Viewer
Developer	Circuits, artifacts, SDK, RPC	Overview, Artifact Manager, Proof Builder, RPC Panel, CLI
Settings	Identity, privacy, RPC, drift, diagnostics	User, Privacy, Network, Prime/Drift, Security, Diagnostics